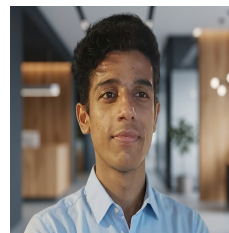


RUPANKAR DUTTA

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🐙 [GitHub](#) 🔗 [LinkedIn](#)



Profile

- B.Tech student at KIIT University specializing in embedded systems, Internet of Things, and edge artificial intelligence, with hands-on experience designing decentralized real-time systems on ESP32. Experienced in developing sensor-integrated firmware, implementing low-latency wireless communication, and deploying lightweight machine learning models for on-device inference. Also proficient in building full-stack applications and system-level solutions combining software, hardware, and data-driven logic.

Education

Kalinga Institute of Industrial Technology (KIIT)
Bachelor of Technology

2023–2027
CGPA: 8.4/10

Skills

- **Programming:** Python, C, C++, JavaScript, SQL
- **Embedded Systems / IoT:** ESP32, Sensor Integration, Embedded Firmware, Edge AI, TinyML
- **Machine Learning:** Facial Recognition, MobileNetV2, LSTM Autoencoders, TensorFlow Lite, Model Quantization
- **Backend Development:** FastAPI, Flask, Node.js, Express, REST APIs, JWT Authentication
- **Web:** React, Socket.IO, MySQL, PostgreSQL, Redis
- **Tools:** Arduino IDE, PlatformIO, EasyEDA, Git, Docker, VS Code

Technologies

- **Embedded:** ESP32-WROOM-32, ESP32-CAM, ESP-NOW, MPU-6050, I2C
- **ML Stack:** Edge Impulse, TensorFlow Lite, TFLite Micro

Projects

- **Intelligent Decentralized Home Automation System (ESP32, Edge AI)** – Developed a fully decentralized smart home system using ESP32 nodes communicating via ESP-NOW for real-time sensing, local decision-making, and offline automation without cloud dependency. Implemented embedded firmware for DHT11, MQ-135, PIR, and sound sensors with filtering and threshold-based alert logic, and built a MobileNetV2-based TinyML facial recognition model for secure on-device access control on ESP32-CAM.
- **Edge AI Anomaly Detection System (ESP32, TinyML)** 🐙 [GitHub](#) – Built a real-time vibration anomaly detection system running entirely on an ESP32 with an MPU-6050 sensor: trained an LSTM autoencoder in TensorFlow, quantized it to int8 for TFLite Micro on-device inference, and added a fault-type classifier. Developed a Flask + JavaScript dashboard with live reconstruction-error charts, adaptive EWMA thresholding, and CSV-exportable anomaly event logs.
- **TheRUPOgate – Microservices API Gateway (Node.js)** 🐙 [GitHub](#) – Engineered a production-style API gateway serving as a single entry point for microservices, handling JWT authentication with role-based header injection, Redis sliding-window rate limiting, and path-prefix proxy routing to backend services. Added structured JSON logging and a live monitoring dashboard streaming request metrics over Server-Sent Events.
- **CookMind AI – AI-Powered Recipe Recommendation Platform** 🐙 [GitHub](#) – Developed a full-stack AI application using React, FastAPI, PostgreSQL, and the OpenAI API delivering personalized recipe recommendations, conversational cooking assistance, and image-based ingredient recognition via Google Vision. Built a modular REST backend with JWT authentication, pantry persistence, user personalization, and chat memory, and containerized the stack with Docker and a CI workflow.
- **Real-Time Multiplayer Quiz Platform** – Designed and built a full-stack multiplayer quiz application using React, Node.js, and Socket.IO for synchronized gameplay across private room-based sessions. Implemented timed rounds, live score tracking, leaderboard generation, and low-latency state synchronization for smooth multi-user gameplay.
- **Automated Plant Watering System (ESP32)** – Built an IoT-based irrigation system using ESP32 and environmental sensing to automate watering decisions based on plant conditions. Integrated real-time monitoring and control features to improve water efficiency and support reliable plant care.